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SUMMARY (cont.)

- If natural frequencies are written as integral multiples of fundamental frequency, then the frequencies are said to be in harmonics. Thus, the first harmonic is $v_1 = v_1$, (the fundamental frequency is called first harmonics), the second harmonics is $v_2 = 2\ v_1$, the third harmonics is $v_3 = 3\ v_1$, and so on.
- *Loudness of sound* is defined as *"the degree of sensation of sound produced in the ear or the perception of sound by the listener"*.
- *The intensity of sound* is defined as *"the sound power transmitted per unit area placed normal to the propagation of sound wave "*.
- Sound intensity level, $\Delta L = 10 \log_{10} \frac{I_1}{I_0}$ *decibel* .
- A closed organ pipe has only odd harmonics and the corresponding frequency of the nth harmonic is $f_n = (2n + 1)\ f_1$.
- In a closed organ pipe the frequencies of harmonics are in the ratio

  $f_1 : f_2 : f_3 : f_4 : \ldots = 1 : 3 : 5 : 7 : \ldots$
- The open organ pipe has all harmonics and frequency of the nth harmonic is $f_n = n\ f_1$.
- In the open organ pipe the frequencies of harmonics are in the ratio

  $f_1 : f_2 : f_3 : f_4 : \ldots = 1 : 2 : 3 : 4 : \ldots$
- When the source and the observer are in relative motion with respect to the medium in which sound propagates, the frequency of the sound wave observed is different from the frequency of the source. This phenomenon is called Doppler Effect. The frequency received by the observer is known as apparent frequency.
- When the Source moves towards a stationary observer, the apparent frequency $f' = f\left(1 + \frac{v_s}{v}\right)$.
- When the Source moves away from a stationary observer, $f' = f\left(1 - \frac{v_s}{v}\right)$.
- When the Observer moves towards the stationary source, $f' = f\left(1 + \frac{v_0}{v}\right)$.
- When the Observer recedes away from the stationary source, $f' = f\left(1 - \frac{v_0}{v}\right)$.
- When the Source and Observer approach each other, $f' = \left(\frac{v + v_0}{v - v_s}\right) f$.
- When the Source and Observer recede from each other, $f' = \left(\frac{v - v_0}{v + v_s}\right) f$.
- When the Source chases the Observer, $f' = \left(\frac{v - v_0}{v - v_s}\right) \mathrm{f}$.
- When the Observer chases the Source, $f' = \left(\frac{v + v_0}{v + v_s}\right) \mathrm{f}$.